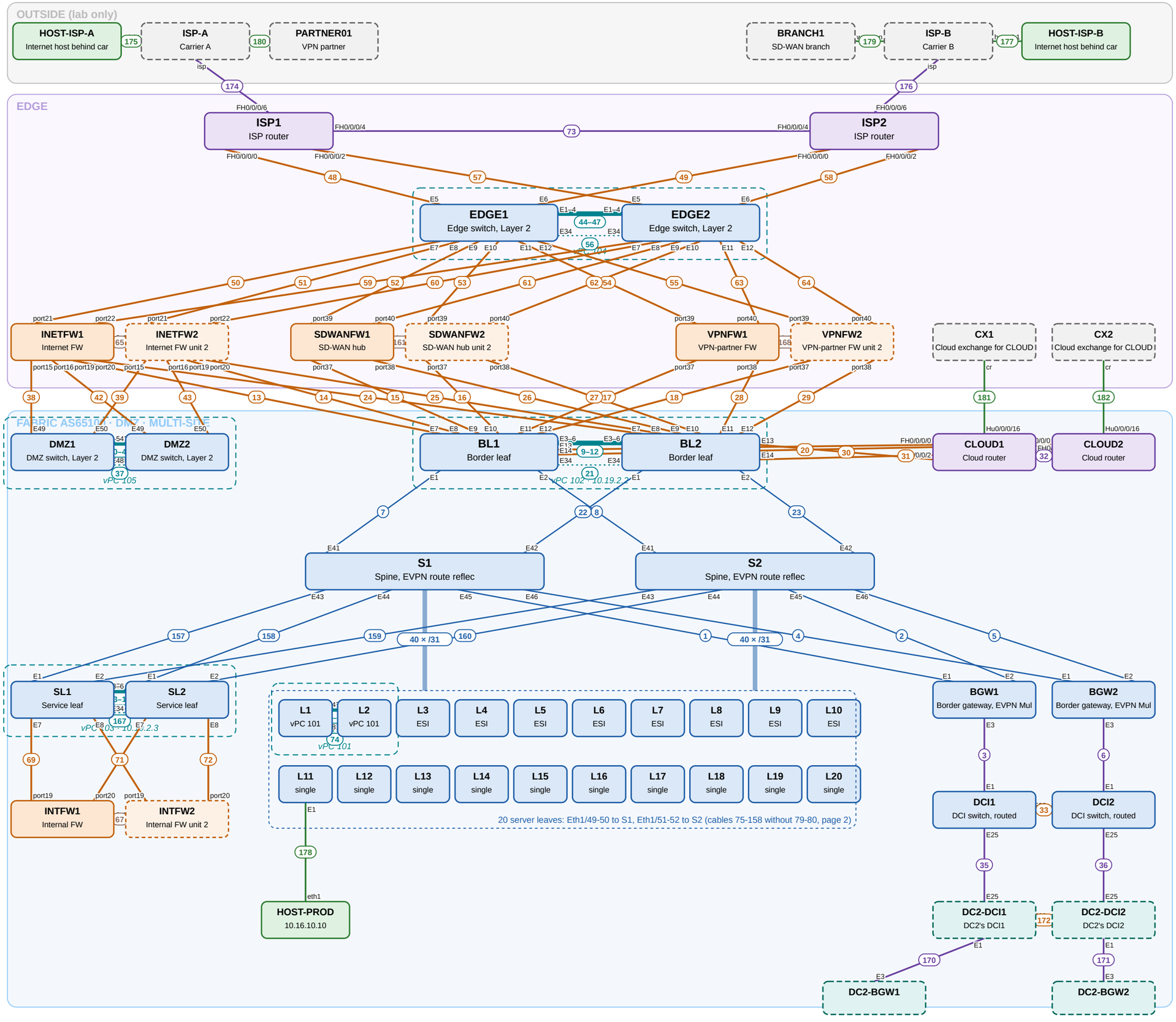
Entire DC1 site: every device and connection



Legend

- Nexus switch: N9Kv 10.6(4)M lite, 8 GB, 3 vCPU
- FortiGate: FRR 10.6.0 stand-in (unit 1 active)
- FortiGate unit 2: stand-in with data ports down
- Cisco 8201 (IOS XR): FRR 10.6.0 stand-in
- DC2 device as FRR stand-in (control plane only)
- Outside node, lab only: FRR 10.6.0
- Test host: Alpine 3.22

- Fabric /31: OSPF UNDERLAY area 0, MTU 9216
- vPC peer-link port-channel1000 (members drawn as one)
- vPC peer-keepalive, VRF VPC-KEEPALIVE /31
- LACP bundle member: 802.1Q trunk
- FortiGate HA heartbeat (ha1/ha2, ha3/ha4)
- Routed link outside the fabric (cross-links, DCI, carriers)
- Lab-only link (hosts, branch, partner, cloud exchange)

- 75 cable number (order of lab-topology-full.DC1.yml; pages 6–8)
- vPC domain (number, shared VTEP)

How to read the overview

One line = one connection; '+' after a port = the first of several cables (all on pages 2-4 and 6-8). Spine to server-leaf cables are drawn as two bundles.

The lab at a glance

Lab vamd-dc1-full · 59 nodes · 182 cables
34 Nexus N9Kv (8 GB each = 272 GB) + 25 FRR / hosts
About 290 GB in total: rendered, not run.
Mgmt network 10.250.0.0/24 (production mgmt0 addresses; lab-only nodes .240-.253).
Covers all 167 cables of DC1's vendor cable schedule.

What stands in for what

FortiGates, Cisco 8201s: FRR, same routing as their rendered FortiOS / IOS XR configs (page 14).
Unit 2 of each FortiGate pair: data ports down (A/P HA shares the addresses).
DC2-*: DC2's DCI and BGW switches, rendered from DC2's own data: Multi-Site control plane.
IPsec (SD-WAN, VPN partner): GRE in the lab.
CX1/CX2: AWS (AS 65500) and Azure (AS 12076) peers of the cloud routers.

Source

lab-topology-full.DC1.yml (branch p5-full-lab), the DC1 configs and cable schedule (main a7bcc19).
Decisions: claude/option5-decisions.md.

Entire DC1: spine port map (S1, S2)

DC1-S1 · lo0 10.19.0.1 · mgmt 10.250.0.2 · 64 × 400G (N9K-C9364D-GX2A): Eth1/1-46 used, Eth1/47-64 spare

1/1	1/2	1/3	1/4	1/5	1/6	1/7	1/8	1/9	1/10	1/11	1/12	1/13	1/14	1/15	1/16	1/17	1/18	1/19	1/20	1/21	1/22	1/23	1/24	1/25	1/26	1/27	1/28	1/29	1/30	1/31	1/32
L1	L1	L2	L2	L3	L3	L4	L4	L5	L5	L6	L6	L7	L7	L8	L8	L9	L9	L10	L10	L11	L11	L12	L12	L13	L13	L14	L14	L15	L15	L16	L16
1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50
.0	.2	.4	.6	.8	.10	.12	.14	.16	.18	.20	.22	.24	.26	.28	.30	.32	.34	.36	.38	.40	.42	.44	.46	.48	.50	.52	.54	.56	.58	.60	.62
#75	#76	#81	#82	#85	#86	#89	#90	#93	#94	#97	#98	#101	#102	#105	#106	#109	#110	#113	#114	#117	#118	#121	#122	#125	#126	#129	#130	#133	#134	#137	#138

1/33	1/34	1/35	1/36	1/37	1/38	1/39	1/40	1/41	1/42	1/43	1/44	1/45	1/46	1/47	1/48	1/49	1/50	1/51	1/52	1/53	1/54	1/55	1/56	1/57	1/58	1/59	1/60	1/61	1/62	1/63	1/64
L17	L17	L18	L18	L19	L19	L20	L20	BL1	BL2	SL1	SL2	BGW1	BGW2	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare
1/49	1/50	1/49	1/50	1/49	1/50	1/49	1/50	1/1	1/1	1/1	1/1	1/1	1/1																		
.64	.66	.68	.70	.72	.74	.76	.78	.80	.82	.84	.86	.88	.90																		
#141	#142	#145	#146	#149	#150	#153	#154	#7	#22	#157	#158	#1	#4																		

Each cell: spine port · far device · far port · spine address of the /31 (10.19.4.x; the far end has x+1) · cable number. Server leaf n uses spine ports 1/(2n-1) and 1/(2n). Every link: OSPF UNDERLAY, iBGP EVPN from loopback0 to the spines.

DC1-S2 · lo0 10.19.0.2 · mgmt 10.250.0.3 · 64 × 400G (N9K-C9364D-GX2A): Eth1/1-46 used, Eth1/47-64 spare

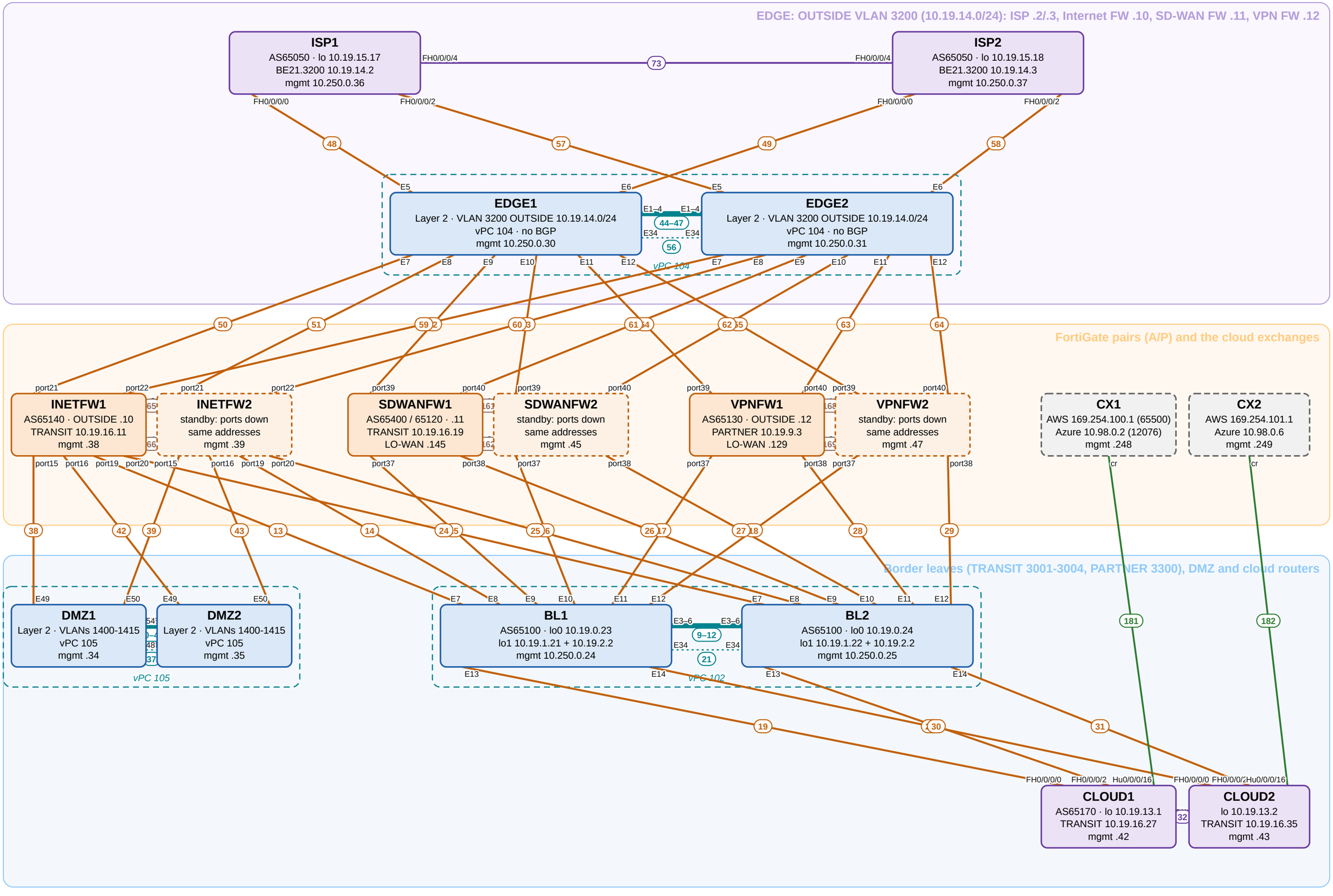
1/1	1/2	1/3	1/4	1/5	1/6	1/7	1/8	1/9	1/10	1/11	1/12	1/13	1/14	1/15	1/16	1/17	1/18	1/19	1/20	1/21	1/22	1/23	1/24	1/25	1/26	1/27	1/28	1/29	1/30	1/31	1/32
L1	L1	L2	L2	L3	L3	L4	L4	L5	L5	L6	L6	L7	L7	L8	L8	L9	L9	L10	L10	L11	L11	L12	L12	L13	L13	L14	L14	L15	L15	L16	L16
1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52
.92	.94	.96	.98	.100	.102	.104	.106	.108	.110	.112	.114	.116	.118	.120	.122	.124	.126	.128	.130	.132	.134	.136	.138	.140	.142	.144	.146	.148	.150	.152	.154
#77	#78	#83	#84	#87	#88	#91	#92	#95	#96	#99	#100	#103	#104	#107	#108	#111	#112	#115	#116	#119	#120	#123	#124	#127	#128	#131	#132	#135	#136	#139	#140

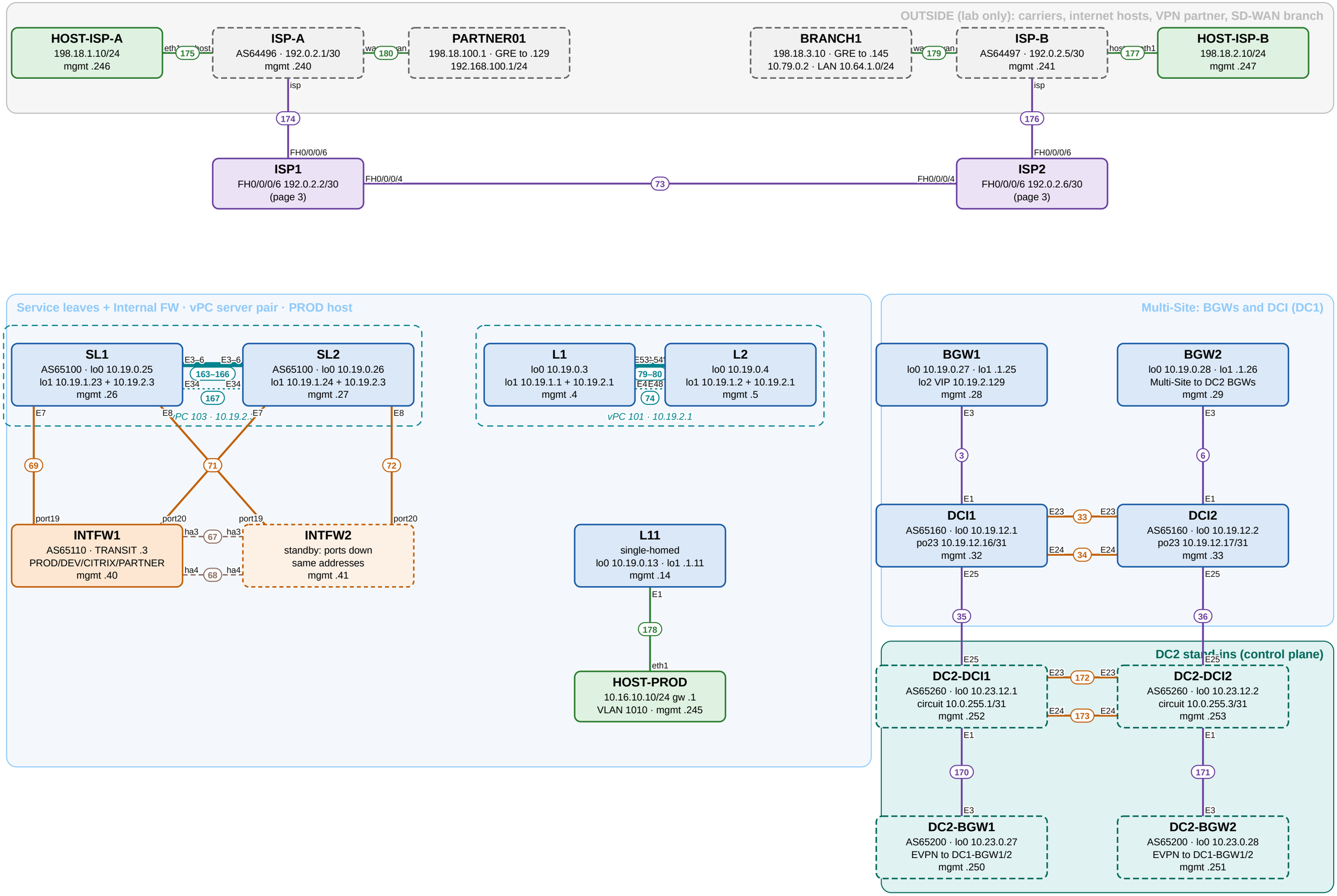
1/33	1/34	1/35	1/36	1/37	1/38	1/39	1/40	1/41	1/42	1/43	1/44	1/45	1/46	1/47	1/48	1/49	1/50	1/51	1/52	1/53	1/54	1/55	1/56	1/57	1/58	1/59	1/60	1/61	1/62	1/63	1/64
L17	L17	L18	L18	L19	L19	L20	L20	BL1	BL2	SL1	SL2	BGW1	BGW2	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare	spare
1/51	1/52	1/51	1/52	1/51	1/52	1/51	1/52	1/2	1/2	1/2	1/2	1/2	1/2																		
.156	.158	.160	.162	.164	.166	.168	.170	.172	.174	.176	.178	.180	.182																		
#143	#144	#147	#148	#151	#152	#155	#156	#8	#23	#159	#160	#2	#5																		

Each cell: spine port · far device · far port · spine address of the /31 (10.19.4.x; the far end has x+1) · cable number. Server leaf n uses spine ports 1/(2n-1) and 1/(2n). Every link: OSPF UNDERLAY, iBGP EVPN from loopback0 to the spines.

server leaf L1-L20 (two ports each) · border leaf · service leaf · border gateway · spare

Server leaves by pairing: L1/L2 vPC domain 101 (shared VTEP 10.19.2.1, peer-link Eth1/53-54, keepalive Eth1/48); L3/L4, L5/L6, L7/L8, L9/L10 ESI pairs (no peer-link; EVPN multihoming is rejected by the virtual image, so they run single-homed in the lab); L11-L20 single-homed. Every server leaf: uplinks Eth1/49-50 to S1 and Eth1/51-52 to S2 (100G in production, 4 x 100G up for 48 x 25G down = 3:1).
Route reflection: each spine has 26 EVPN clients (20 server leaves, BL1/2, SL1/2, BGW1/2), no spine-to-spine session.
Spine port map rule (generator): leaf n -> spine ports 1/(2n-1), 1/(2n); 1/41-42 BL1/BL2; 1/43-44 SL1/SL2; 1/45-46 BGW1/BGW2; 1/47-64 spare (growth to 29 leaves).


EDGE: OUTSIDE VLAN 3200 (10.19.14.0/24): ISP .2/.3, Internet FW .10, SD-WAN FW .11, VPN FW .12
ISP1
AS65050 · lo 10.19.15.17
BE21.3200 10.19.14.2
mgmt 10.250.0.36
ISP2
AS65050 · lo 10.19.15.18
BE21.3200 10.19.14.3
mgmt 10.250.0.37
FH0/0/0/4
73
FH0/0/0/0
FH0/0/0/2
48
57
49
58
EDGE1
Layer 2 · VLAN 3200 OUTSIDE 10.19.14.0/24
vPC 104 · no BGP
mgmt 10.250.0.30
EDGE2
Layer 2 · VLAN 3200 OUTSIDE 10.19.14.0/24
vPC 104 · no BGP
mgmt 10.250.0.31
E1–4
44–47
E34
56
E5
E6
E7
E8
E9
E10
E11
E12
vPC 104
FortiGate pairs (A/P) and the cloud exchanges
INETFW1
AS65140 · OUTSIDE .10
TRANSIT 10.19.16.11
mgmt .38
INETFW2
standby: ports down
same addresses
mgmt .39
SDWANFW1
AS65400 / 65120 · .11
TRANSIT 10.19.16.19
LO-WAN .145
SDWANFW2
standby: ports down
same addresses
mgmt .45
VPNFW1
AS65130 · OUTSIDE .12
PARTNER 10.19.9.3
LO-WAN .129
VPNFW2
standby: ports down
same addresses
mgmt .47
port21
port22
port39
port40
port15
port16
port19
port20
port37
port38
CX1
AWS 169.254.100.1 (65500)
Azure 10.98.0.2 (12076)
mgmt .248
CX2
AWS 169.254.101.1
Azure 10.98.0.6
mgmt .249
cr
181
182
50
51
59
60
61
62
63
64
38
42
39
43
13
14
24
25
26
27
28
29
Border leaves (TRANSIT 3001-3004, PARTNER 3300), DMZ and cloud routers
DMZ1
Layer 2 · VLANs 1400-1415
vPC 105
mgmt .34
DMZ2
Layer 2 · VLANs 1400-1415
vPC 105
mgmt .35
vPC 105
E49
E50
BL1
AS65100 · lo0 10.19.0.23
lo1 10.19.1.21 + 10.19.2.2
mgmt 10.250.0.24
BL2
AS65100 · lo0 10.19.0.24
lo1 10.19.1.22 + 10.19.2.2
mgmt 10.250.0.25
E3–6
9–12
21
E13
E14
vPC 102
19
30
31
CLOUD1
AS65170 · lo 10.19.13.1
TRANSIT 10.19.16.27
mgmt .42
CLOUD2
lo 10.19.13.2
TRANSIT 10.19.16.35
mgmt .43
32
FH0/0/0/0
FH0/0/0/2
Hu0/0/0/16


OUTSIDE (lab only): carriers, internet hosts, VPN partner, SD-WAN branch
HOST-ISP-A
198.18.1.10/24
mgmt .246
175
ISP-A
AS64496 · 192.0.2.1/30
mgmt .240
180
PARTNER01
198.18.100.1 · GRE to .129
192.168.100.1/24
BRANCH1
198.18.3.10 · GRE to .145
10.79.0.2 · LAN 10.64.1.0/24
179
ISP-B
AS64497 · 192.0.2.5/30
mgmt .241
177
HOST-ISP-B
198.18.2.10/24
mgmt .247
isp
174
176
FH0/0/0/6
ISP1
FH0/0/0/6 192.0.2.2/30
(page 3)
FH0/0/0/4
73
ISP2
FH0/0/0/6 192.0.2.6/30
(page 3)
Service leaves + Internal FW · vPC server pair · PROD host
SL1
AS65100 · lo0 10.19.0.25
lo1 10.19.1.23 + 10.19.2.3
mgmt .26
E3–6
163–166
E34
167
SL2
AS65100 · lo0 10.19.0.26
lo1 10.19.1.24 + 10.19.2.3
mgmt .27
vPC 103 · 10.19.2.3
E7
E8
69
71
72
port19
port20
INTFW1
AS65110 · TRANSIT .3
PROD/DEV/CITRIX/PARTNER
mgmt .40
ha3
67
ha4
68
INTFW2
standby: ports down
same addresses
mgmt .41
L1
lo0 10.19.0.3
lo1 10.19.1.1 + 10.19.2.1
mgmt .4
E53–54
79–80
E4E48
74
L2
lo0 10.19.0.4
lo1 10.19.1.2 + 10.19.2.1
mgmt .5
vPC 101 · 10.19.2.1
L11
single-homed
lo0 10.19.0.13 · lo1 .1.11
mgmt .14
E1
178
eth1
HOST-PROD
10.16.10.10/24 gw .1
VLAN 1010 · mgmt .245
Multi-Site: BGWs and DCI (DC1)
BGW1
lo0 10.19.0.27 · lo1 .1.25
lo2 VIP 10.19.2.129
mgmt .28
BGW2
lo0 10.19.0.28 · lo1 .1.26
Multi-Site to DC2 BGWs
mgmt .29
E3
3
6
DCI1
AS65160 · lo0 10.19.12.1
po23 10.19.12.16/31
mgmt .32
E23
33
E24
34
DCI2
AS65160 · lo0 10.19.12.2
po23 10.19.12.17/31
mgmt .33
E25
35
36
DC2 stand-ins (control plane)
DC2-DCI1
AS65260 · lo0 10.23.12.1
circuit 10.0.255.1/31
mgmt .252
172
173
DC2-DCI2
AS65260 · lo0 10.23.12.2
circuit 10.0.255.3/31
mgmt .253
170
171
DC2-BGW1
AS65200 · lo0 10.23.0.27
EVPN to DC1-BGW1/2
mgmt .250
DC2-BGW2
AS65200 · lo0 10.23.0.28
EVPN to DC1-BGW1/2
mgmt .251

Entire DC1: the 59 lab nodes

Node	Stands for / role	Lab image	BGP AS	loopback0 / main address	loopback1 (VTEP) / second address	Pair	mgmt
DC1-S1	Spine, EVPN route reflector (N9K-C9364D-GX2A)	N9Kv 8 GB	65100	10.19.0.1	-		10.250.0.2
DC1-S2	Spine, EVPN route reflector (N9K-C9364D-GX2A)	N9Kv 8 GB	65100	10.19.0.2	-		10.250.0.3
DC1-L1	Server leaf	N9Kv 8 GB	65100	10.19.0.3	10.19.1.1 + 10.19.2.1	vPC 101 (with L2)	10.250.0.4
DC1-L2	Server leaf	N9Kv 8 GB	65100	10.19.0.4	10.19.1.2 + 10.19.2.1	vPC 101 (with L1)	10.250.0.5
DC1-L3	Server leaf	N9Kv 8 GB	65100	10.19.0.5	10.19.1.3	ESI pair L3/L4	10.250.0.6
DC1-L4	Server leaf	N9Kv 8 GB	65100	10.19.0.6	10.19.1.4	ESI pair L3/L4	10.250.0.7
DC1-L5	Server leaf	N9Kv 8 GB	65100	10.19.0.7	10.19.1.5	ESI pair L5/L6	10.250.0.8
DC1-L6	Server leaf	N9Kv 8 GB	65100	10.19.0.8	10.19.1.6	ESI pair L5/L6	10.250.0.9
DC1-L7	Server leaf	N9Kv 8 GB	65100	10.19.0.9	10.19.1.7	ESI pair L7/L8	10.250.0.10
DC1-L8	Server leaf	N9Kv 8 GB	65100	10.19.0.10	10.19.1.8	ESI pair L7/L8	10.250.0.11
DC1-L9	Server leaf	N9Kv 8 GB	65100	10.19.0.11	10.19.1.9	ESI pair L9/L10	10.250.0.12
DC1-L10	Server leaf	N9Kv 8 GB	65100	10.19.0.12	10.19.1.10	ESI pair L9/L10	10.250.0.13
DC1-L11	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.13	10.19.1.11	single-homed	10.250.0.14
DC1-L12	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.14	10.19.1.12	single-homed	10.250.0.15
DC1-L13	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.15	10.19.1.13	single-homed	10.250.0.16
DC1-L14	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.16	10.19.1.14	single-homed	10.250.0.17
DC1-L15	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.17	10.19.1.15	single-homed	10.250.0.18
DC1-L16	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.18	10.19.1.16	single-homed	10.250.0.19
DC1-L17	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.19	10.19.1.17	single-homed	10.250.0.20
DC1-L18	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.20	10.19.1.18	single-homed	10.250.0.21
DC1-L19	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.21	10.19.1.19	single-homed	10.250.0.22
DC1-L20	Server leaf (single-homed)	N9Kv 8 GB	65100	10.19.0.22	10.19.1.20	single-homed	10.250.0.23
DC1-BL1	Border leaf	N9Kv 8 GB	65100	10.19.0.23	10.19.1.21 + 10.19.2.2	vPC 102	10.250.0.24
DC1-BL2	Border leaf	N9Kv 8 GB	65100	10.19.0.24	10.19.1.22 + 10.19.2.2	vPC 102	10.250.0.25
DC1-SL1	Service leaf	N9Kv 8 GB	65100	10.19.0.25	10.19.1.23 + 10.19.2.3	vPC 103	10.250.0.26
DC1-SL2	Service leaf	N9Kv 8 GB	65100	10.19.0.26	10.19.1.24 + 10.19.2.3	vPC 103	10.250.0.27
DC1-BGW1	Border gateway, EVPN Multi-Site	N9Kv 8 GB	65100	10.19.0.27	10.19.1.25; lo2 VIP 10.19.2.129	anycast BGW	10.250.0.28
DC1-BGW2	Border gateway, EVPN Multi-Site	N9Kv 8 GB	65100	10.19.0.28	10.19.1.26; lo2 VIP 10.19.2.129	anycast BGW	10.250.0.29
DC1-DCI1	DCI switch, routed	N9Kv 8 GB	65160	10.19.12.1	-	pair po23	10.250.0.32
DC1-DCI2	DCI switch, routed	N9Kv 8 GB	65160	10.19.12.2	-	pair po23	10.250.0.33
DC1-EDGE1	Edge switch, Layer 2 (VLAN 3200)	N9Kv 8 GB	-	-	-	vPC 104	10.250.0.30
DC1-EDGE2	Edge switch, Layer 2 (VLAN 3200)	N9Kv 8 GB	-	-	-	vPC 104	10.250.0.31
DC1-DMZ1	DMZ switch, Layer 2 (VLANs 1400-1415)	N9Kv 8 GB	-	-	-	vPC 105	10.250.0.34
DC1-DMZ2	DMZ switch, Layer 2 (VLANs 1400-1415)	N9Kv 8 GB	-	-	-	vPC 105	10.250.0.35
DC1-INTFW1	Internal FW (FortiGate) unit 1	FRR 10.6.0	65110	-	-	HA A/P	10.250.0.40
DC1-INTFW2	Internal FW unit 2 (standby)	FRR 10.6.0	65110	-	-	HA A/P	10.250.0.41
DC1-INETFW1	Internet FW (FortiGate) unit 1	FRR 10.6.0	65140	-	-	HA A/P	10.250.0.38
DC1-INETFW2	Internet FW unit 2 (standby)	FRR 10.6.0	65140	-	-	HA A/P	10.250.0.39
DC1-SDWANFW1	SD-WAN hub (FortiGate) unit 1	FRR 10.6.0	65400 (65120 to the BLs)	LO-WAN 198.51.100.145	hub tunnel 10.79.0.1/21	HA A/P	10.250.0.44
DC1-SDWANFW2	SD-WAN hub unit 2 (standby)	FRR 10.6.0	65400 (65120 to the BLs)	LO-WAN 198.51.100.145	-	HA A/P	10.250.0.45
DC1-VPNFW1	VPN-partner FW (FortiGate) unit 1	FRR 10.6.0	65130	LO-WAN 198.51.100.129	-	HA A/P	10.250.0.46
DC1-VPNFW2	VPN-partner FW unit 2 (standby)	FRR 10.6.0	65130	LO-WAN 198.51.100.129	-	HA A/P	10.250.0.47
DC1-ISP1	ISP router (Cisco 8201)	FRR 10.6.0	65050	10.19.15.17	-		10.250.0.36
DC1-ISP2	ISP router (Cisco 8201)	FRR 10.6.0	65050	10.19.15.18	-		10.250.0.37
DC1-CLOUD1	Cloud router (Cisco 8201)	FRR 10.6.0	65170	10.19.13.1	-		10.250.0.42
DC1-CLOUD2	Cloud router (Cisco 8201)	FRR 10.6.0	65170	10.19.13.2	-		10.250.0.43
DC2-DCI1	DC2's DCI1 (stand-in)	FRR 10.6.0	65260	10.23.12.1	-	lab only	10.250.0.252
DC2-DCI2	DC2's DCI2 (stand-in)	FRR 10.6.0	65260	10.23.12.2	-	lab only	10.250.0.253
DC2-BGW1	DC2's BGW1 (stand-in, control plane)	FRR 10.6.0	65200	10.23.0.27	10.23.1.25; lo2 10.23.2.129	lab only	10.250.0.250
DC2-BGW2	DC2's BGW2 (stand-in, control plane)	FRR 10.6.0	65200	10.23.0.28	10.23.1.26	lab only	10.250.0.251
DC1-ISP-A	Carrier A (lab only)	FRR 10.6.0	64496	-	-		10.250.0.240
DC1-ISP-B	Carrier B (lab only)	FRR 10.6.0	64497	-	-		10.250.0.241
DC1-BRANCH1	SD-WAN branch (lab only)	FRR 10.6.0	65400	LAN 10.64.1.1/24	tunnel 10.79.0.2		10.250.0.242
DC1-PARTNER01	VPN partner (lab only)	FRR 10.6.0	-	198.18.100.1	protected 192.168.100.1/24		10.250.0.243
DC1-CX1	Cloud exchange for CLOUD1: AWS + Azure (lab only)	FRR 10.6.0	65500 / 12076	172.16.1.1 (aws)	172.30.1.1 (azure)		10.250.0.248
DC1-CX2	Cloud exchange for CLOUD2: AWS + Azure (lab only)	FRR 10.6.0	65500 / 12076	172.16.1.1 (aws)	172.30.1.1 (azure)		10.250.0.249
DC1-HOST-PROD	PROD test host on L11 Eth1/1	Alpine 3.22	-	10.16.10.10/24	gw 10.16.10.1		10.250.0.245
DC1-HOST-ISP-A	Internet host behind carrier A	Alpine 3.22	-	198.18.1.10/24	gw .1		10.250.0.246
DC1-HOST-ISP-B	Internet host behind carrier B	Alpine 3.22	-	198.18.2.10/24	gw .1		10.250.0.247

Server leaf n: loopback0 10.19.0.(n+2), loopback1 10.19.1.n, mgmt 10.250.0.(n+3). vPC pair L1/L2 shares 10.19.2.1; ESI pairs (L3/L4 … L9/L10) have no peer-link (EVPN multihoming: `evpn esi multihoming` is rejected by the virtual image, so ESI is not active in the lab). Values: the DC1 production configs.

Entire DC1: cables 1-61 of 182

#	End A (port)	End B (port)	Type	A side: address / VLANs / port-channel	B side	Media	Length
1	DC1-BGW1 **E1**	DC1-S1 **E45**	fabric /31	10.19.4.89/31	10.19.4.88/31	dac	2 m
2	DC1-BGW1 **E2**	DC1-S2 **E45**	fabric /31	10.19.4.181/31	10.19.4.180/31	smf	26 m
3	DC1-BGW1 **E3**	DC1-DCI1 **E1**	routed	10.19.12.33/31	10.19.12.32/31	dac	2 m
4	DC1-BGW2 **E1**	DC1-S1 **E46**	fabric /31	10.19.4.91/31	10.19.4.90/31	smf	26 m
5	DC1-BGW2 **E2**	DC1-S2 **E46**	fabric /31	10.19.4.183/31	10.19.4.182/31	dac	2 m
6	DC1-BGW2 **E3**	DC1-DCI2 **E1**	routed	10.19.12.35/31	10.19.12.34/31	dac	2 m
7	DC1-BL1 **E1**	DC1-S1 **E41**	fabric /31	10.19.4.81/31	10.19.4.80/31	dac	2 m
8	DC1-BL1 **E2**	DC1-S2 **E41**	fabric /31	10.19.4.173/31	10.19.4.172/31	smf	26 m
9	DC1-BL1 **E3**	DC1-BL2 **E3**	vPC peer-link	vlan 2531,2599,3001-3004,3300,3900 po 1000	vlan 2531,2599,3001-3004,3300,3900 po 1000	smf	26 m
10	DC1-BL1 **E4**	DC1-BL2 **E4**	vPC peer-link	vlan 2531,2599,3001-3004,3300,3900 po 1000	vlan 2531,2599,3001-3004,3300,3900 po 1000	smf	26 m
11	DC1-BL1 **E5**	DC1-BL2 **E5**	vPC peer-link	vlan 2531,2599,3001-3004,3300,3900 po 1000	vlan 2531,2599,3001-3004,3300,3900 po 1000	smf	26 m
12	DC1-BL1 **E6**	DC1-BL2 **E6**	vPC peer-link	vlan 2531,2599,3001-3004,3300,3900 po 1000	vlan 2531,2599,3001-3004,3300,3900 po 1000	smf	26 m
13	DC1-BL1 **E7**	DC1-INETFW1 **port19**	LACP member	vlan 3001 po 11		dac	3 m
14	DC1-BL1 **E8**	DC1-INETFW2 **port19**	LACP member	vlan 3001 po 12		smf	27 m
15	DC1-BL1 **E9**	DC1-SDWANFW1 **port37**	LACP member	vlan 3002 po 13		dac	3 m
16	DC1-BL1 **E10**	DC1-SDWANFW2 **port37**	LACP member	vlan 3002 po 14		smf	27 m
17	DC1-BL1 **E11**	DC1-VPNFW1 **port37**	LACP member	vlan 3300 po 15		dac	2 m
18	DC1-BL1 **E12**	DC1-VPNFW2 **port37**	LACP member	vlan 3300 po 16		smf	27 m
19	DC1-BL1 **E13**	DC1-CLOUD1 **FH0/0/0/0**	LACP member	vlan 3003 po 31	bundle 31	dac	2 m
20	DC1-BL1 **E14**	DC1-CLOUD2 **FH0/0/0/0**	LACP member	vlan 3004 po 32	bundle 31	smf	27 m
21	DC1-BL1 **E34**	DC1-BL2 **E34**	vPC keepalive	vrf VPC-KEEPALIVE 10.19.17.2/31	vrf VPC-KEEPALIVE 10.19.17.3/31	smf	26 m
22	DC1-BL2 **E1**	DC1-S1 **E42**	fabric /31	10.19.4.83/31	10.19.4.82/31	smf	26 m
23	DC1-BL2 **E2**	DC1-S2 **E42**	fabric /31	10.19.4.175/31	10.19.4.174/31	dac	2 m
24	DC1-BL2 **E7**	DC1-INETFW1 **port20**	LACP member	vlan 3001 po 11		smf	27 m
25	DC1-BL2 **E8**	DC1-INETFW2 **port20**	LACP member	vlan 3001 po 12		dac	3 m
26	DC1-BL2 **E9**	DC1-SDWANFW1 **port38**	LACP member	vlan 3002 po 13		smf	27 m
27	DC1-BL2 **E10**	DC1-SDWANFW2 **port38**	LACP member	vlan 3002 po 14		dac	3 m
28	DC1-BL2 **E11**	DC1-VPNFW1 **port38**	LACP member	vlan 3300 po 15		smf	27 m
29	DC1-BL2 **E12**	DC1-VPNFW2 **port38**	LACP member	vlan 3300 po 16		dac	2 m
30	DC1-BL2 **E13**	DC1-CLOUD1 **FH0/0/0/2**	LACP member	vlan 3003 po 31	bundle 31	smf	27 m
31	DC1-BL2 **E14**	DC1-CLOUD2 **FH0/0/0/2**	LACP member	vlan 3004 po 32	bundle 31	dac	2 m
32	DC1-CLOUD1 **FH0/0/0/4**	DC1-CLOUD2 **FH0/0/0/4**	routed	10.19.13.16/31	10.19.13.17/31	smf	28 m
33	DC1-DCI1 **E23**	DC1-DCI2 **E23**	LACP member	po 23	po 23	smf	26 m
34	DC1-DCI1 **E24**	DC1-DCI2 **E24**	LACP member	po 23	po 23	smf	26 m
35	DC1-DCI1 **E25**	DC2-DCI1 **E25**	routed	10.0.255.0/31	10.0.255.1/31		
36	DC1-DCI2 **E25**	DC2-DCI2 **E25**	routed	10.0.255.2/31	10.0.255.3/31		
37	DC1-DMZ1 **E48**	DC1-DMZ2 **E48**	vPC keepalive	vrf VPC-KEEPALIVE 10.19.17.8/31	vrf VPC-KEEPALIVE 10.19.17.9/31	smf	26 m
38	DC1-DMZ1 **E49**	DC1-INETFW1 **port15**	LACP member	vlan 1400-1415 po 11		dac	2 m
39	DC1-DMZ1 **E50**	DC1-INETFW2 **port15**	LACP member	vlan 1400-1415 po 12		smf	27 m
40	DC1-DMZ1 **E53**	DC1-DMZ2 **E53**	vPC peer-link	vlan 1400-1415 po 1000	vlan 1400-1415 po 1000	smf	26 m
41	DC1-DMZ1 **E54**	DC1-DMZ2 **E54**	vPC peer-link	vlan 1400-1415 po 1000	vlan 1400-1415 po 1000	smf	26 m
42	DC1-DMZ2 **E49**	DC1-INETFW1 **port16**	LACP member	vlan 1400-1415 po 11		smf	27 m
43	DC1-DMZ2 **E50**	DC1-INETFW2 **port16**	LACP member	vlan 1400-1415 po 12		dac	2 m
44	DC1-EDGE1 **E1**	DC1-EDGE2 **E1**	vPC peer-link	vlan 3200 po 1000	vlan 3200 po 1000	smf	26 m
45	DC1-EDGE1 **E2**	DC1-EDGE2 **E2**	vPC peer-link	vlan 3200 po 1000	vlan 3200 po 1000	smf	26 m
46	DC1-EDGE1 **E3**	DC1-EDGE2 **E3**	vPC peer-link	vlan 3200 po 1000	vlan 3200 po 1000	smf	26 m
47	DC1-EDGE1 **E4**	DC1-EDGE2 **E4**	vPC peer-link	vlan 3200 po 1000	vlan 3200 po 1000	smf	26 m
48	DC1-EDGE1 **E5**	DC1-ISP1 **FH0/0/0/0**	LACP member	vlan 3200 po 21	bundle 21	dac	2 m
49	DC1-EDGE1 **E6**	DC1-ISP2 **FH0/0/0/0**	LACP member	vlan 3200 po 22	bundle 21	smf	26 m
50	DC1-EDGE1 **E7**	DC1-INETFW1 **port21**	LACP member	vlan 3200 po 11		dac	3 m
51	DC1-EDGE1 **E8**	DC1-INETFW2 **port21**	LACP member	vlan 3200 po 12		smf	27 m
52	DC1-EDGE1 **E9**	DC1-SDWANFW1 **port39**	LACP member	vlan 3200 po 13		dac	2 m
53	DC1-EDGE1 **E10**	DC1-SDWANFW2 **port39**	LACP member	vlan 3200 po 14		smf	27 m
54	DC1-EDGE1 **E11**	DC1-VPNFW1 **port39**	LACP member	vlan 3200 po 15		dac	2 m
55	DC1-EDGE1 **E12**	DC1-VPNFW2 **port39**	LACP member	vlan 3200 po 16		smf	27 m
56	DC1-EDGE1 **E34**	DC1-EDGE2 **E34**	vPC keepalive	vrf VPC-KEEPALIVE 10.19.17.6/31	vrf VPC-KEEPALIVE 10.19.17.7/31	smf	26 m
57	DC1-EDGE2 **E5**	DC1-ISP1 **FH0/0/0/2**	LACP member	vlan 3200 po 21	bundle 21	smf	26 m
58	DC1-EDGE2 **E6**	DC1-ISP2 **FH0/0/0/2**	LACP member	vlan 3200 po 22	bundle 21	dac	2 m
59	DC1-EDGE2 **E7**	DC1-INETFW1 **port22**	LACP member	vlan 3200 po 11		smf	27 m
60	DC1-EDGE2 **E8**	DC1-INETFW2 **port22**	LACP member	vlan 3200 po 12		dac	3 m
61	DC1-EDGE2 **E9**	DC1-SDWANFW1 **port40**	LACP member	vlan 3200 po 13		smf	27 m

Ports: E*n* = Ethernet1/*n* (containerlab eth*n*); FH0/0/0/*2k* = lab Gi0-0-0-*k*; Hu0/0/0/16 = lab Hu0-0-0-16; FortiGate ports by their FortiOS names. Media and length: the DC1 vendor cable schedule (main a7bcc19). Numbers = the order of lab-topology-full.DC1.yml.

Entire DC1: cables 62-122 of 182

#	End A (port)	End B (port)	Type	A side: address / VLANs / port-channel	B side	Media	Length
62	DC1-EDGE2 **E10**	DC1-SDWANFW2 **port40**	LACP member	vlan 3200 po 14		dac	2 m
63	DC1-EDGE2 **E11**	DC1-VPNFW1 **port40**	LACP member	vlan 3200 po 15		smf	27 m
64	DC1-EDGE2 **E12**	DC1-VPNFW2 **port40**	LACP member	vlan 3200 po 16		dac	2 m
65	DC1-INETFW1 **ha3**	DC1-INETFW2 **ha3**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
66	DC1-INETFW1 **ha4**	DC1-INETFW2 **ha4**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
67	DC1-INTFW1 **ha3**	DC1-INTFW2 **ha3**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
68	DC1-INTFW1 **ha4**	DC1-INTFW2 **ha4**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
69	DC1-INTFW1 **port19**	DC1-SL1 **E7**	LACP member		vlan 3000,3100-3101,3105-3106 po 11	dac	3 m
70	DC1-INTFW1 **port20**	DC1-SL2 **E7**	LACP member		vlan 3000,3100-3101,3105-3106 po 11	smf	27 m
71	DC1-INTFW2 **port19**	DC1-SL1 **E8**	LACP member		vlan 3000,3100-3101,3105-3106 po 12	smf	27 m
72	DC1-INTFW2 **port20**	DC1-SL2 **E8**	LACP member		vlan 3000,3100-3101,3105-3106 po 12	dac	3 m
73	DC1-ISP1 **FH0/0/0/4**	DC1-ISP2 **FH0/0/0/4**	routed	10.19.15.0/31	10.19.15.1/31	smf	26 m
74	DC1-L1 **E48**	DC1-L2 **E48**	vPC keepalive	vrf VPC-KEEPALIVE 10.19.17.0/31	vrf VPC-KEEPALIVE 10.19.17.1/31	smf	35 m
75	DC1-L1 **E49**	DC1-S1 **E1**	fabric /31	10.19.4.1/31	10.19.4.0/31	smf	28 m
76	DC1-L1 **E50**	DC1-S1 **E2**	fabric /31	10.19.4.3/31	10.19.4.2/31	smf	28 m
77	DC1-L1 **E51**	DC1-S2 **E1**	fabric /31	10.19.4.93/31	10.19.4.92/31	smf	31 m
78	DC1-L1 **E52**	DC1-S2 **E2**	fabric /31	10.19.4.95/31	10.19.4.94/31	smf	31 m
79	DC1-L1 **E53**	DC1-L2 **E53**	vPC peer-link	vlan 1010-1017,1040-1051,1110-1118,1140-1150,1500-1501,2501-2502,2522,3900 po 1000	same	smf	35 m
80	DC1-L1 **E54**	DC1-L2 **E54**	vPC peer-link	vlan 1010-1017,1040-1051,1110-1118,1140-1150,1500-1501,2501-2502,2522,3900 po 1000	same	smf	35 m
81	DC1-L2 **E49**	DC1-S1 **E3**	fabric /31	10.19.4.5/31	10.19.4.4/31	smf	28 m
82	DC1-L2 **E50**	DC1-S1 **E4**	fabric /31	10.19.4.7/31	10.19.4.6/31	smf	28 m
83	DC1-L2 **E51**	DC1-S2 **E3**	fabric /31	10.19.4.97/31	10.19.4.96/31	smf	31 m
84	DC1-L2 **E52**	DC1-S2 **E4**	fabric /31	10.19.4.99/31	10.19.4.98/31	smf	31 m
85	DC1-L3 **E49**	DC1-S1 **E5**	fabric /31	10.19.4.9/31	10.19.4.8/31	smf	28 m
86	DC1-L3 **E50**	DC1-S1 **E6**	fabric /31	10.19.4.11/31	10.19.4.10/31	smf	28 m
87	DC1-L3 **E51**	DC1-S2 **E5**	fabric /31	10.19.4.101/31	10.19.4.100/31	smf	31 m
88	DC1-L3 **E52**	DC1-S2 **E6**	fabric /31	10.19.4.103/31	10.19.4.102/31	smf	31 m
89	DC1-L4 **E49**	DC1-S1 **E7**	fabric /31	10.19.4.13/31	10.19.4.12/31	smf	23 m
90	DC1-L4 **E50**	DC1-S1 **E8**	fabric /31	10.19.4.15/31	10.19.4.14/31	smf	23 m
91	DC1-L4 **E51**	DC1-S2 **E7**	fabric /31	10.19.4.105/31	10.19.4.104/31	smf	31 m
92	DC1-L4 **E52**	DC1-S2 **E8**	fabric /31	10.19.4.107/31	10.19.4.106/31	smf	31 m
93	DC1-L5 **E49**	DC1-S1 **E9**	fabric /31	10.19.4.17/31	10.19.4.16/31	smf	23 m
94	DC1-L5 **E50**	DC1-S1 **E10**	fabric /31	10.19.4.19/31	10.19.4.18/31	smf	23 m
95	DC1-L5 **E51**	DC1-S2 **E9**	fabric /31	10.19.4.109/31	10.19.4.108/31	smf	26 m
96	DC1-L5 **E52**	DC1-S2 **E10**	fabric /31	10.19.4.111/31	10.19.4.110/31	smf	26 m
97	DC1-L6 **E49**	DC1-S1 **E11**	fabric /31	10.19.4.21/31	10.19.4.20/31	smf	26 m
98	DC1-L6 **E50**	DC1-S1 **E12**	fabric /31	10.19.4.23/31	10.19.4.22/31	smf	26 m
99	DC1-L6 **E51**	DC1-S2 **E11**	fabric /31	10.19.4.113/31	10.19.4.112/31	smf	26 m
100	DC1-L6 **E52**	DC1-S2 **E12**	fabric /31	10.19.4.115/31	10.19.4.114/31	smf	26 m
101	DC1-L7 **E49**	DC1-S1 **E13**	fabric /31	10.19.4.25/31	10.19.4.24/31	smf	26 m
102	DC1-L7 **E50**	DC1-S1 **E14**	fabric /31	10.19.4.27/31	10.19.4.26/31	smf	26 m
103	DC1-L7 **E51**	DC1-S2 **E13**	fabric /31	10.19.4.117/31	10.19.4.116/31	smf	23 m
104	DC1-L7 **E52**	DC1-S2 **E14**	fabric /31	10.19.4.119/31	10.19.4.118/31	smf	23 m
105	DC1-L8 **E49**	DC1-S1 **E15**	fabric /31	10.19.4.29/31	10.19.4.28/31	smf	31 m
106	DC1-L8 **E50**	DC1-S1 **E16**	fabric /31	10.19.4.31/31	10.19.4.30/31	smf	31 m
107	DC1-L8 **E51**	DC1-S2 **E15**	fabric /31	10.19.4.121/31	10.19.4.120/31	smf	23 m
108	DC1-L8 **E52**	DC1-S2 **E16**	fabric /31	10.19.4.123/31	10.19.4.122/31	smf	23 m
109	DC1-L9 **E49**	DC1-S1 **E17**	fabric /31	10.19.4.33/31	10.19.4.32/31	smf	31 m
110	DC1-L9 **E50**	DC1-S1 **E18**	fabric /31	10.19.4.35/31	10.19.4.34/31	smf	31 m
111	DC1-L9 **E51**	DC1-S2 **E17**	fabric /31	10.19.4.125/31	10.19.4.124/31	smf	28 m
112	DC1-L9 **E52**	DC1-S2 **E18**	fabric /31	10.19.4.127/31	10.19.4.126/31	smf	28 m
113	DC1-L10 **E49**	DC1-S1 **E19**	fabric /31	10.19.4.37/31	10.19.4.36/31	smf	31 m
114	DC1-L10 **E50**	DC1-S1 **E20**	fabric /31	10.19.4.39/31	10.19.4.38/31	smf	31 m
115	DC1-L10 **E51**	DC1-S2 **E19**	fabric /31	10.19.4.129/31	10.19.4.128/31	smf	28 m
116	DC1-L10 **E52**	DC1-S2 **E20**	fabric /31	10.19.4.131/31	10.19.4.130/31	smf	28 m
117	DC1-L11 **E49**	DC1-S1 **E21**	fabric /31	10.19.4.41/31	10.19.4.40/31	smf	28 m
118	DC1-L11 **E50**	DC1-S1 **E22**	fabric /31	10.19.4.43/31	10.19.4.42/31	smf	28 m
119	DC1-L11 **E51**	DC1-S2 **E21**	fabric /31	10.19.4.133/31	10.19.4.132/31	smf	31 m
120	DC1-L11 **E52**	DC1-S2 **E22**	fabric /31	10.19.4.135/31	10.19.4.134/31	smf	31 m
121	DC1-L12 **E49**	DC1-S1 **E23**	fabric /31	10.19.4.45/31	10.19.4.44/31	smf	28 m
122	DC1-L12 **E50**	DC1-S1 **E24**	fabric /31	10.19.4.47/31	10.19.4.46/31	smf	28 m

Ports: E*n* = Ethernet1/*n* (containerlab eth*n*); FH0/0/0/*2k* = lab Gi0-0-0-*k*; Hu0/0/0/16 = lab Hu0-0-0-16; FortiGate ports by their FortiOS names. Media and length: the DC1 vendor cable schedule (main a7bcc19). Numbers = the order of lab-topology-full.DC1.yml.

Entire DC1: cables 123-182 of 182

#	End A (port)	End B (port)	Type	A side: address / VLANs / port-channel	B side	Media	Length
123	DC1-L12 **E51**	DC1-S2 **E23**	fabric /31	10.19.4.137/31	10.19.4.136/31	smf	31 m
124	DC1-L12 **E52**	DC1-S2 **E24**	fabric /31	10.19.4.139/31	10.19.4.138/31	smf	31 m
125	DC1-L13 **E49**	DC1-S1 **E25**	fabric /31	10.19.4.49/31	10.19.4.48/31	smf	28 m
126	DC1-L13 **E50**	DC1-S1 **E26**	fabric /31	10.19.4.51/31	10.19.4.50/31	smf	28 m
127	DC1-L13 **E51**	DC1-S2 **E25**	fabric /31	10.19.4.141/31	10.19.4.140/31	smf	31 m
128	DC1-L13 **E52**	DC1-S2 **E26**	fabric /31	10.19.4.143/31	10.19.4.142/31	smf	31 m
129	DC1-L14 **E49**	DC1-S1 **E27**	fabric /31	10.19.4.53/31	10.19.4.52/31	smf	23 m
130	DC1-L14 **E50**	DC1-S1 **E28**	fabric /31	10.19.4.55/31	10.19.4.54/31	smf	23 m
131	DC1-L14 **E51**	DC1-S2 **E27**	fabric /31	10.19.4.145/31	10.19.4.144/31	smf	31 m
132	DC1-L14 **E52**	DC1-S2 **E28**	fabric /31	10.19.4.147/31	10.19.4.146/31	smf	31 m
133	DC1-L15 **E49**	DC1-S1 **E29**	fabric /31	10.19.4.57/31	10.19.4.56/31	smf	23 m
134	DC1-L15 **E50**	DC1-S1 **E30**	fabric /31	10.19.4.59/31	10.19.4.58/31	smf	23 m
135	DC1-L15 **E51**	DC1-S2 **E29**	fabric /31	10.19.4.149/31	10.19.4.148/31	smf	26 m
136	DC1-L15 **E52**	DC1-S2 **E30**	fabric /31	10.19.4.151/31	10.19.4.150/31	smf	26 m
137	DC1-L16 **E49**	DC1-S1 **E31**	fabric /31	10.19.4.61/31	10.19.4.60/31	smf	26 m
138	DC1-L16 **E50**	DC1-S1 **E32**	fabric /31	10.19.4.63/31	10.19.4.62/31	smf	26 m
139	DC1-L16 **E51**	DC1-S2 **E31**	fabric /31	10.19.4.153/31	10.19.4.152/31	smf	26 m
140	DC1-L16 **E52**	DC1-S2 **E32**	fabric /31	10.19.4.155/31	10.19.4.154/31	smf	26 m
141	DC1-L17 **E49**	DC1-S1 **E33**	fabric /31	10.19.4.65/31	10.19.4.64/31	smf	26 m
142	DC1-L17 **E50**	DC1-S1 **E34**	fabric /31	10.19.4.67/31	10.19.4.66/31	smf	26 m
143	DC1-L17 **E51**	DC1-S2 **E33**	fabric /31	10.19.4.157/31	10.19.4.156/31	smf	23 m
144	DC1-L17 **E52**	DC1-S2 **E34**	fabric /31	10.19.4.159/31	10.19.4.158/31	smf	23 m
145	DC1-L18 **E49**	DC1-S1 **E35**	fabric /31	10.19.4.69/31	10.19.4.68/31	smf	31 m
146	DC1-L18 **E50**	DC1-S1 **E36**	fabric /31	10.19.4.71/31	10.19.4.70/31	smf	31 m
147	DC1-L18 **E51**	DC1-S2 **E35**	fabric /31	10.19.4.161/31	10.19.4.160/31	smf	23 m
148	DC1-L18 **E52**	DC1-S2 **E36**	fabric /31	10.19.4.163/31	10.19.4.162/31	smf	23 m
149	DC1-L19 **E49**	DC1-S1 **E37**	fabric /31	10.19.4.73/31	10.19.4.72/31	smf	31 m
150	DC1-L19 **E50**	DC1-S1 **E38**	fabric /31	10.19.4.75/31	10.19.4.74/31	smf	31 m
151	DC1-L19 **E51**	DC1-S2 **E37**	fabric /31	10.19.4.165/31	10.19.4.164/31	smf	28 m
152	DC1-L19 **E52**	DC1-S2 **E38**	fabric /31	10.19.4.167/31	10.19.4.166/31	smf	28 m
153	DC1-L20 **E49**	DC1-S1 **E39**	fabric /31	10.19.4.77/31	10.19.4.76/31	smf	31 m
154	DC1-L20 **E50**	DC1-S1 **E40**	fabric /31	10.19.4.79/31	10.19.4.78/31	smf	31 m
155	DC1-L20 **E51**	DC1-S2 **E39**	fabric /31	10.19.4.169/31	10.19.4.168/31	smf	28 m
156	DC1-L20 **E52**	DC1-S2 **E40**	fabric /31	10.19.4.171/31	10.19.4.170/31	smf	28 m
157	DC1-S1 **E43**	DC1-SL1 **E1**	fabric /31	10.19.4.84/31	10.19.4.85/31	dac	2 m
158	DC1-S1 **E44**	DC1-SL2 **E1**	fabric /31	10.19.4.86/31	10.19.4.87/31	smf	26 m
159	DC1-S2 **E43**	DC1-SL1 **E2**	fabric /31	10.19.4.176/31	10.19.4.177/31	smf	26 m
160	DC1-S2 **E44**	DC1-SL2 **E2**	fabric /31	10.19.4.178/31	10.19.4.179/31	dac	2 m
161	DC1-SDWANFW1 **ha1**	DC1-SDWANFW2 **ha1**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
162	DC1-SDWANFW1 **ha2**	DC1-SDWANFW2 **ha2**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
163	DC1-SL1 **E3**	DC1-SL2 **E3**	vPC peer-link	vlan 2501-2502,2522,2531,2599,3000,3100-3101,3105-3106,3900 po 1000	same	smf	26 m
164	DC1-SL1 **E4**	DC1-SL2 **E4**	vPC peer-link	vlan 2501-2502,2522,2531,2599,3000,3100-3101,3105-3106,3900 po 1000	same	smf	26 m
165	DC1-SL1 **E5**	DC1-SL2 **E5**	vPC peer-link	vlan 2501-2502,2522,2531,2599,3000,3100-3101,3105-3106,3900 po 1000	same	smf	26 m
166	DC1-SL1 **E6**	DC1-SL2 **E6**	vPC peer-link	vlan 2501-2502,2522,2531,2599,3000,3100-3101,3105-3106,3900 po 1000	same	smf	26 m
167	DC1-SL1 **E34**	DC1-SL2 **E34**	vPC keepalive	vrf VPC-KEEPALIVE 10.19.17.4/31	vrf VPC-KEEPALIVE 10.19.17.5/31	smf	26 m
168	DC1-VPNFW1 **ha1**	DC1-VPNFW2 **ha1**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
169	DC1-VPNFW1 **ha2**	DC1-VPNFW2 **ha2**	FW HA	HA heartbeat	HA heartbeat	smf	28 m
170	DC2-BGW1 **E3**	DC2-DCI1 **E1**	routed	10.23.12.33/31	10.23.12.32/31	lab	DC2 stand-ins
171	DC2-BGW2 **E3**	DC2-DCI2 **E1**	routed	10.23.12.35/31	10.23.12.34/31	lab	DC2 stand-ins
172	DC2-DCI1 **E23**	DC2-DCI2 **E23**	LACP member	po 23 (10.23.12.16/31)	po 23 (10.23.12.17/31)	lab	DC2 stand-ins
173	DC2-DCI1 **E24**	DC2-DCI2 **E24**	LACP member	po 23	po 23	lab	DC2 stand-ins
174	DC1-ISP1 **FH0/0/0/6**	DC1-ISP-A **isp**	routed	192.0.2.2/30	192.0.2.1/30	carrier	lab only
175	DC1-ISP-A **host**	DC1-HOST-ISP-A **eth1**	lab only	198.18.1.1/24	198.18.1.10/24	lab	lab only
176	DC1-ISP2 **FH0/0/0/6**	DC1-ISP-B **isp**	routed	192.0.2.6/30	192.0.2.5/30	carrier	lab only
177	DC1-ISP-B **host**	DC1-HOST-ISP-B **eth1**	lab only	198.18.2.1/24	198.18.2.10/24	lab	lab only
178	DC1-L11 **E1**	DC1-HOST-PROD **eth1**	lab only	access vlan 1010	10.16.10.10/24	lab	lab only (P3-39)
179	DC1-ISP-B **wan**	DC1-BRANCH1 **wan**	lab only	198.18.3.254/24	198.18.3.10/24	lab	lab only
180	DC1-ISP-A **wan**	DC1-PARTNER01 **wan**	lab only	198.18.100.254/24	198.18.100.1/24	lab	lab only
181	DC1-CLOUD1 **Hu0/0/0/16**	DC1-CX1 **cr**	lab only	.100 169.254.100.2/30, .200 10.98.0.1/30	.100 169.254.100.1/30, .200 10.98.0.2/30	cloud xc	lab only
182	DC1-CLOUD2 **Hu0/0/0/16**	DC1-CX2 **cr**	lab only	.101 169.254.101.2/30, .200 10.98.0.5/30	.101 169.254.101.1/30, .200 10.98.0.6/30	cloud xc	lab only

Ports: E*n* = Ethernet1/*n* (containerlab eth*n*); FH0/0/0/*2k* = lab Gi0-0-0-*k*; Hu0/0/0/16 = lab Hu0-0-0-16; FortiGate ports by their FortiOS names. Media and length: the DC1 vendor cable schedule (main a7bcc19). Numbers = the order of lab-topology-full.DC1.yml.

Entire DC1: BGP sessions

Session	Who (address)	AS	Notes
iBGP EVPN (l2vpn evpn)	26 clients → S1 10.19.0.1, S2 10.19.0.2	65100	L1-L20, BL1/2, SL1/2, BGW1/2 from loopback0; spines are route reflectors; BFD multihop 999 × 3
eBGP TRANSIT	BL1 .9 / BL2 .10 ↔ INETFW1 .11	65100 ↔ 65140	VLAN 3001, 10.19.16.8/29
eBGP TRANSIT	BL1 .17 / BL2 .18 ↔ SDWANFW1 .19	65100 ↔ 65120	VLAN 3002, 10.19.16.16/29; the hub runs AS 65400 with local-as 65120 toward the leaves
eBGP TRANSIT	BL1 .25 / BL2 .26 ↔ CLOUD1 .27	65100 ↔ 65170	VLAN 3003, 10.19.16.24/29
eBGP TRANSIT	BL1 .33 / BL2 .34 ↔ CLOUD2 .35	65100 ↔ 65170	VLAN 3004, 10.19.16.32/29
eBGP PARTNER	BL1 10.19.9.1 / BL2 .2 ↔ VPNFW1 .3	65100 ↔ 65130	VLAN 3300; the FW announces the partner's translated /28 10.16.160.0/28
eBGP 5 tenant VRFs	SL1 / SL2 ↔ INTFW1	65100 ↔ 65110	TRANSIT 3000, PROD 3100, DEV 3101, CITRIX 3105, PARTNER 3106 (as the lab path)
eBGP OUTSIDE	INETFW1 .10 ↔ ISP1 .2 / ISP2 .3	65140 ↔ 65050	VLAN 3200 through EDGE1/2
iBGP ISP pair / eBGP carriers	ISP1 ↔ ISP2; ISP1 ↔ ISP-A, ISP2 ↔ ISP-B	65050; 64496, 64497	as the lab path
iBGP cloud pair	CLOUD1 10.19.13.16 ↔ CLOUD2 .17	65170	cross-link FH0/0/0/4, next-hop-self
eBGP AWS	CLOUD1 169.254.100.2 ↔ CX1 .1; CLOUD2 169.254.101.2 ↔ CX2 .1	65170 ↔ 65500	VLANs 100/101 on Hu0/0/0/16; in: 172.16.0.0/13 le 24; out: 0/0, 10.16.0.0/16, PARTNER summary
eBGP Azure	CLOUD1 10.98.0.1 ↔ CX1 .2; CLOUD2 10.98.0.5 ↔ CX2 .6	65170 ↔ 12076	VLAN 200; in: 172.30.0.0/15 le 24; BFD 300 ms × 3
iBGP overlay (SD-WAN)	SDWANFW1 10.79.0.1 ↔ BRANCH1 10.79.0.2	65400	dynamic neighbours on 10.79.0.0/21 (hub = route reflector); GRE in the lab
eBGP underlay to DCI	BGW1 .33 ↔ DCI1 .32; BGW2 .35 ↔ DCI2 .34 (10.19.12.x)	65100 ↔ 65160	loopbacks advertised toward DC2
iBGP DCI pair	DCI1 10.19.12.16 ↔ DCI2 .17 (po23)	65160	next-hop-self
eBGP inter-site circuits	DCI1 10.0.255.0 ↔ DC2-DCI1 .1; DCI2 .2 ↔ DC2-DCI2 .3	65160 ↔ 65260	Ethernet1/25 (vendor circuits)
eBGP Multi-Site EVPN	BGW1/BGW2 ↔ DC2-BGW1 10.23.0.27 / DC2-BGW2 .28	65100 ↔ 65200	loopback0 to loopback0, multihop, peer-type fabric-external, rewrite-evpn-rt-asn

No BGP to the VPN partner: the VPN FW routes 192.168.100.0/24 into the tunnel statically. Every session: password `cisco123` (public lab constant). Values: the DC1 rendered configs; the CX and branch sides: the option 5 stand-ins. Expected in a running lab and not seen yet: every session above except those the 20 Oct laptop smoke test brought up without a Nexus (page 15).

Entire DC1: overlay (VRFs, VNIs, VTEPs)

VRFs

VRF	L3 VNI	L3VNI VLAN	Tenant VLANs (L2 VNI = 10000 + VLAN)	Where	FW legs
PROD	50001	2501	1010-1017, 1040-1051	L1-L20 (server leaves), SL1/2, BGW1/2 (multi-site)	3100 → INTFW vrf10
DEV	50002	2502	1110-1118, 1140-1150	L1-L20, SL1/2, BGW1/2 (multi-site)	3101 → INTFW vrf11
CITRIX	50022	2522	1500-1501	L1-L20, SL1/2	3105 → INTFW vrf15
PARTNER	50031	2531	-	SL1/2, BL1/2	3106 → INTFW vrf16; BL 3300 → VPN FW
TRANSIT	50100	2599	-	SL1/2, BL1/2	3000 → INTFW vrf1; BL 3001 INETFW, 3002 SD-WAN, 3003/3004 clouds
REPLICATION	50041	2541	1598	BGW1/2 (multi-site)	-

VTEPs (loopback1)

Switches	Own VTEP address	Shared address
L1, L2 (vPC 101)	10.19.1.1, 10.19.1.2	10.19.2.1
L3-L10 (ESI pairs)	10.19.1.3 - 10.19.1.10	-
L11-L20 (single-homed)	10.19.1.11 - 10.19.1.20	-
BL1, BL2 (vPC 102)	10.19.1.21, 10.19.1.22	10.19.2.2
SL1, SL2 (vPC 103)	10.19.1.23, 10.19.1.24	10.19.2.3
BGW1, BGW2	10.19.1.25, 10.19.1.26	Multi-Site VIP 10.19.2.129 (loopback2)

27 VTEPs in the site (20 server leaves, 2 border, 2 service, 2 BGWs, plus the shared addresses). BUM: ingress replication over BGP (EVPN type-3); the BGWs replicate between the sites (`multisite ingress-replication`). The VLAN → VNI → VRF table of the whole site is in the DC1 site tables (diagrams/DC1-tables.md). Values: the DC1 rendered configs.

Entire DC1: packet walk, PROD server to AWS

#	Device	Lookup	Sends it	Cables
1	**DC1-HOST-PROD**	10.16.10.10 → 172.16.1.1 (AWS, on CX1)	default gateway 10.16.10.1 on L11 (anycast, MAC 0200.0000.00aa), VLAN 1010	cable 178
2	**DC1-L11**	VRF PROD: 0.0.0.0/0 from the service leaves (EVPN type-5)	VXLAN L3 VNI 50001, outer 10.19.1.11 → SL1 or SL2's own VTEP (advertise-pip)	117-120
3	**DC1-S1 / S2**	underlay only (OSPF)	outer IP routed, inner untouched	157-160
4	**DC1-SL1 / SL2**	VRF PROD: 0/0 via the Internal FW 10.19.8.3	VLAN 3100 on port-channel 11	69-72
5	**DC1-INTFW1**	vrf10 has only 0/0 (leaked from vrf1); vrf1 has only 0/0 from the SLs	back to the SL pair on VLAN 3000 (TRANSIT)	69-70
6	**DC1-SL1 / SL2**	VRF TRANSIT: 172.16.1.0/24 from the border leaves (EVPN, VNI 50100); longer than 0/0	VXLAN to BL1 or BL2's own VTEP	157-160, 7-8, 22-23
7	**DC1-BL1 / BL2**	VRF TRANSIT: 172.16.1.0/24 via CLOUD1 10.19.16.27 (eBGP 65170)	VLAN 3003 on port-channel 31	19, 30
8	**DC1-CLOUD1**	172.16.1.0/24 from AWS 169.254.100.1 (RP-AWS-IN: 172.16.0.0/13 le 24)	out of Hu0/0/0/16.100	181
9	**DC1-CX1 (AWS side)**	VRF aws: 172.16.1.1 is local (dummy interface)	reply to 10.16.10.10: CX1 has 10.16.0.0/16, the cloud router's aggregate	

Why the packet passes the Internal FW twice

The tenant VRFs only get 0/0 from the Internal FW (route-map RM-LEAK-TRANSIT-TO-PROD matches PL-DEFAULT), so PROD has no cloud route of its own: the FW sees and can inspect every PROD → cloud flow, then hands it to TRANSIT, where the border leaves know the cloud /24s from the cloud routers.

Return

CX1 (AWS) → CLOUD1, which sends 10.16.0.0/16 (aggregate) to AWS and learns 10.16.10.0/24 … from the border leaves (RP-FABRIC-IN 10.16.0.0/16 le 24) → BL (TRANSIT) → EVPN → SL → INTFW (vrf1 → vrf10, tenant summary) → SL (VRF PROD; host route 10.16.10.10/32 is an EVPN type-2 from L11) → L11 → host.

Other paths worth walking in this lab

From → to	Path
PROD host → internet host A	as the DC1 lab path (page 4 there): SL → INTFW → TRANSIT → BL → INETFW (NAT 198.51.100.1-126) → EDGE → ISP1 → carrier A
SD-WAN branch LAN 10.64.1.0/24 → PROD	BRANCH1 → GRE over carrier B → ISP2 → EDGE (VLAN 3200) → SDWANFW1 LO-WAN 198.51.100.145 → vrf1 → BL (TRANSIT 3002) → … → INTFW → PROD
VPN partner 192.168.100.0/24 → PARTNER VRF	PARTNER01 → GRE over carrier A → ISP1 → EDGE → VPNFW1 LO-WAN 198.51.100.129 → vrf16 (SNAT to 10.16.160.0/28) → BL (PARTNER 3300)
DC1 PROD ↔ DC2 PROD (multi-site VRF)	L11 → spines → BGW1/2 (re-originated, Multi-Site VIP 10.19.2.129) → DCI → circuit → DC2 (stand-ins: control plane only, no data plane)

Not run: this page is the design read from the rendered configs and the stand-ins. Check with the commands on the DC1 lab path PDF pages 4–5.

Entire DC1: the first ARP across leaves

Step	What happens	Where to look
1	A server on DC1-L11 (VLAN 1010) ARPs for its gateway 10.16.10.1: L11 answers itself (anycast gateway 0200.0000.00aa).	`show ip arp vrf PROD` on L11
2	L11 sends one EVPN type-2 (MAC + IP, L2 VNI 11010, L3 VNI 50001, next hop 10.19.1.11) to S1 and S2.	`show bgp l2vpn evpn 10.16.10.10`
3	The spines reflect it to the 25 other VTEPs. Every server leaf has VLAN 1010, so all 19 other server leaves install the MAC and the ARP entry (suppression cache); the SL pair imports the /32 into VRF PROD; the BGWs re-originate it toward DC2 (PROD is multi-site).	`show l2route evpn mac-ip all`
4	A server on DC1-L12 ARPs for 10.16.10.10: L12 answers from its suppression cache. The request never leaves L12.	`show ip arp suppression-cache detail` on L12
5	An address nobody has learned: the ARP is copied to every VTEP with VNI 11010 (ingress replication list from EVPN type-3: 20 leaves + BGWs).	`show nve vni 11010`, `show l2route evpn imet all`
6	vPC pair L1/L2: a server dual-homed to both learns on either switch; the type-2 carries the shared VTEP 10.19.2.1. ESI pairs: multihoming is not active on the virtual image (`evpn esi multihoming` rejected), so the ESI leaves behave as single-homed in the lab.	`show vpc brief`, `show nve peers`

This lab has one test host (on L11), so steps 3–5 need a second host on another leaf to be seen: give a server leaf port an access segment in Infrahub (as Eth1/1 on L11, P3-39) and add a host to the topology. Not done here: it would change a production config (D5-36).

The 14 lab-path tests, on the full site

#	Test	Breaks (cable numbers of this lab)	Difference in the full lab
1	Carrier loss	cables 174, 176	as the lab path
2	ISP router down	ISP1	as the lab path
3	Spine down	S1	now 26 leaves and BGWs re-converge on S2
5	Leaf-spine link down	cable 117 (L11 Eth1/49)	
6	vPC peer-link down	BL pair (cables 9-12)	also L1/L2 (79-80), SL (163-166), EDGE (44-47), DMZ (40-41)
7	vPC keepalive down	cable 21	
8	Internal FW bundle member down	cable 69	
9	Silent failure toward the Internal FW	cables 69, 70	
10	Internal FW down	INTFW1	now: swap to INTFW2 by bringing its ports up (HA stand-in)
11	Internet FW down	INETFW1	same, INETFW2
12	Filter test	INTFW1	
13	BGP graceful restart	L11, ISP1	
14	ISP1 returns	ISP1	
15	Node reload with its config	L11	

Tests only this lab can run

#	Test	Breaks	Expected
16	Cloud router down	CLOUD1	TRANSIT keeps 172.16.1.0/24 via CLOUD2; CX1 sessions down
17	AWS VIF down	CLOUD1 Hu0/0/0/16.100	AWS prefixes via CLOUD2's VIF 2
18	Inter-site circuit down	cable 35 (DCI1 Eth1/25)	Multi-Site sessions stay up through DCI2
19	BGW down	BGW1	anycast BGW: BGW2 keeps the VIP 10.19.2.129
20	SD-WAN hub failover	SDWANFW1 → SDWANFW2	bring unit 2's ports up, unit 1's down; branch sessions come back
21	VPN FW failover	VPNFW1 → VPNFW2	same for the partner tunnel

Tests 16–21 are not in `lab/failtests.py` yet. Last results of tests 1–15 (DC1 lab path, AWS 8 Oct): DC1 lab path PDF page 9. None of these has run on the full site.

Entire DC1: what stands in for what

Device	Stand-in (mirrors the rendered config)	Lab-only replacement / not mirrored
Internal / Internet FW (FortiGate)	existing lab-path templates	unit 2: same script, data ports set down by the topology
SD-WAN hub (FortiGate)	frr_sdwan_fw_lab.j2: VRF 0 WAN + VRF 1 TRANSIT/overlay, AS 65400 with local-as 65120, dynamic branch neighbours (RR), prefix-lists, route-maps, aggregates	ADVPN IPsec → GRE to BRANCH1; static 0/0 → ISP1 .2 (no VRRP on FRR)
VPN-partner FW (FortiGate)	frr_vpn_fw_lab.j2: VRF 0 WAN + VRF 16 PARTNER, BGP 65130 to the BLs, translated /28, statics to the partner networks	IKEv2 → GRE to PARTNER01; partner SNAT → iptables (skipped with a warning when iptables cannot be installed)
Cloud router (Cisco 8201)	frr_cloud_router_lab.j2: bundle + TRANSIT sub-interface, cross-link iBGP, AWS and Azure handoffs, all prefix-sets and route-policies	management VRF not mirrored; HundredGigE0/0/0/16 = Hu0-0-0-16
ISP router (Cisco 8201)	existing lab-path template	VRRP not mirrored
DC2 DCI and BGW	frr_dci_lab.j2, frr_bgw_lab.j2, rendered from DC2's own data	control plane only: no VXLAN data plane on the BGW stand-ins
Carriers	existing template + optional port 'wan' (carrier A: 198.18.100.0/24, carrier B: 198.18.3.0/24)	lab only
SD-WAN branch	frr_sdwan_branch_lab.j2 from the hub's data: GRE to LO-WAN, iBGP 65400 to 10.79.0.1, LAN 10.64.1.0/24	lab only
VPN partner	frr_vpn_partner_lab.j2 from the VPN FW's data: peer 198.18.100.1, protected 192.168.100.0/24	lab only
Cloud exchange	frr_cloud_exchange_lab.j2 from the cloud router's data: VRF aws (65500) and azure (12076), 172.16.1.1 / 172.30.1.1 answer pings	lab only

Every stand-in is an Infrahub artifact rendered from the same GraphQL query as the device's production config, so addresses, ASNs, neighbours and policies come from the graph. All decisions behind them (marked R, open for review): `claude/option5-decisions.md`.

What you need

Item	Value
Memory	About 290 GB: 34 Nexus × 8 GB = 272 GB plus 25 FRR / Alpine containers. Not run.
CPU	34 × 3 vCPU for the Nexus guests (102 vCPU), KVM on the host
Time	Not measured; the 9-Nexus lab path took 27 minutes
Artifacts	lab-topology-full, frr-lab-init (FWs, ISP/cloud routers, DC2 DCI/BGW), carrier-, branch-, partner-, cx-lab-init

Bring it up

```
.venv/bin/python lab/labctl.py fetch DC1-full --compare-baselines
.venv/bin/python lab/labctl.py validate DC1-full
.venv/bin/python lab/labctl.py deploy DC1-full
.venv/bin/python lab/labctl.py push DC1-full
.venv/bin/python lab/labctl.py carve DC1-full
```

Take it down

```
.venv/bin/python lab/labctl.py destroy DC1-full
```

Log in

Kind	How
Nexus (34)	`ssh vamdadmin@<mgmt>` (page 5), password from `ztp/make-secret.sh`
FRR (22)	`docker exec -it clab-vamd-dc1-full-DC1-SDWANFW1 vtysh`
Hosts (3)	`docker exec -it clab-vamd-dc1-full-DC1-HOST-PROD sh`

Checked on the laptop (20 Oct, FRR-only, no Nexus)

Check (job 391b, 22 FRR stand-ins + 3 hosts, no Nexus)	Result
Every stand-in init script ends "ready"	22 of 22
ISP pair iBGP, ISP1-carrier A, ISP2-carrier B	4 of 4 Established
Cloud pair iBGP; CLOUD1/CLOUD2 to AWS and Azure (CX1, CX2), both sides	8 of 8 Established
DC2 stand-ins: DCI pair iBGP over po23, DCI-BGW eBGP	6 of 6 Established
CLOUD1 accepts 172.16.1.0/24 (AWS) and 172.30.1.0/24 (Azure)	2 of 2
Pings: CLOUD1/2 to the AWS and Azure addresses; ISP1/2 to the internet hosts	5 of 5
Carrier wan ports reach the VPN partner and the SD-WAN branch	2 of 2
FortiGate unit 2 ports set down by the topology	4 of 4 units
Total	27 pass, 0 fail

Not testable without the Nexus: every session to a leaf, the GRE tunnels (they cross EDGE), Multi-Site EVPN between the DC1 BGWs and the DC2 stand-ins. First run (391) found the VPN FW stand-in failing when iptables could not be installed (no package mirror on the laptop): fixed, NAT skipped with a warning.

FortiGate failover by hand

```
# standby unit up, active unit's data ports down (example: SD-WAN hub)
docker exec clab-vamd-dc1-full-DC1-SDWANFW2 sh -c 'for p in port37 port38 port39 port40; do ip
link set $p up; done'
docker exec clab-vamd-dc1-full-DC1-SDWANFW1 sh -c 'for p in port37 port38 port39 port40; do ip
link set $p down; done'
```

Stop / start one device

As the lab path (LAB-GUI.md): FRR stand-ins and hosts can be stopped and started on their own; a Nexus cannot be started again with `containerlab start --node` (19 Oct test): rebuild the lab.